

Title: Sentiment Analysis of Movie Reviews Using the Stanford IMDB Dataset

Abstract

This project presents a sentiment analysis system built on the Stanford IMDB movie review dataset. The task was to classify reviews as positive or negative using deep learning models. The workflow included text preprocessing, tokenization, padding, model training, and evaluation using accuracy, precision, recall, and F1-score.grammarly+1

Introduction

Sentiment analysis is a common natural language processing task that identifies the emotional tone of text. In this project, movie reviews from the Stanford IMDB dataset were used to train models that predict whether a review is positive or negative. The goal was to compare different neural network architectures and identify the best-performing model.citewrite.qut.edu+1

Dataset Description

The Stanford IMDB dataset contains movie reviews organized into training and testing folders with positive and negative labels. After extraction, the dataset is typically arranged as `train/pos`, `train/neg`, `test/pos`, and `test/neg`. Each review is stored as a text file, making it suitable for supervised text classification.grammarly+1

Methodology

The project began by loading reviews from the dataset folders and assigning binary labels. The text was cleaned by lowercasing and removing unnecessary symbols. A tokenizer converted words into integer sequences, and padding was applied so that all reviews had equal length before feeding them into neural network models.citewrite.qut.edu+1

Three models were used for comparison: 1D CNN, BiLSTM, and GRU. Each model was trained on the training set and validated on a hold-out validation set. Early stopping and checkpointing were used to improve training efficiency and preserve the best model.grammarly+1

Results and Discussion

The models were evaluated using classification metrics such as accuracy, precision, recall, and F1-score. A confusion matrix was also used to examine correct and incorrect predictions. Recurrent models such as BiLSTM and GRU generally perform well on sentiment tasks because they capture word order and sequence information more effectively than simpler models.citewrite.qut.edu+1

Conclusion

This project demonstrated that the Stanford IMDB dataset can be used effectively for binary sentiment classification. By applying text preprocessing and deep learning models, the project achieved a complete end-to-end machine learning workflow. Model comparison helped identify the most suitable architecture for the task.grammarly+1

Future Work

Future improvements may include using pretrained word embeddings, attention

mechanisms, and hyperparameter tuning. The model can also be deployed in a simple web interface for live sentiment prediction.citewrite.qut.edu+1

Short conclusion

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